# Mingjun Wang

Ph.D. Student, Computer Science and Engineering

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## Research Profile

My research focuses on **LLM4EDA / AI4EDA** and **3D IC design automation**. I develop multi-modal circuit representations, cross-stage timing models, and LLM-based verification and optimization methods, alongside legality-aware RTL-to-GDS flows for hybrid-bonded 3D ICs. My background in parallel fault simulation and industrial EDA connects learning-based methods with physical implementation, functional correctness, and reproducible evaluation.

## Education

**The Chinese University of Hong Kong** 2025 – Present

Ph.D. in Computer Science and Engineering | Expected June 2029

Advisor: Prof. Bei Yu.

**University of Chinese Academy of Sciences & ICT, CAS** 2022 – 2025

M.Eng. in Electronic Information | GPA 3.77/4.0 (Top 5%)

Advisors: Prof. Xiaowei Li and Prof. Huawei Li, State Key Laboratory of Processors.

**Peking University** 2018 – 2022

B.Eng. in Microelectronics Science and Technology

Undergraduate thesis: True Random Number Generator Based on RRAM.

## Research and Engineering Experience

### LLM4EDA / AI4EDA and 3D IC Design Automation

May 2024 – Present

Circuit learning, tool-assisted design, and physical implementation

- **Circuit learning and data.** Developed multi-modal representations and cross-stage timing/power models through MOSS (DAC 2025), RTLDistil (ICML 2025 Spotlight), CircuitNet 3.0 (ICLR 2026), and Distill3D (ICCAD 2026). CircuitNet 3.0 provides over 15,000 instances from 8,659 validated designs.
- **LLM-based EDA.** Designed and implemented functional coverage generation methods and tool-based validation in an industry-academic project; contributed to ChatFCM and knowledge-constrained library tuning with LibPilot (ICCAD 2026).
- **3D IC implementation.** Developed a research RTL-to-GDS flow spanning hybrid-bonding structures, clock-tree synthesis, routing, parasitic extraction, and timing analysis; CU-3D is an invited paper at ASP-DAC 2027.
- **Current focus.** Long-horizon LLM agents for chip implementation: tool-feedback-driven decisions, fixed-budget evaluation, and independent post-route verification that distinguishes valid execution from actual PPA improvement.

### Fault Simulation Research

Jul. 2023 – Mar. 2025

State Key Laboratory of Processors, ICT, CAS

- Designed and implemented parallel fault simulation using dominator-tree partitioning, strongly connected components, and pattern parallelism (DomSim, EPICS, and ETPG).
- DomSim achieved a 10.29× average speedup over its commercial baseline; EPICS achieved 5.94× with unchanged

fault coverage, in their respective published experiments (TCAD 2026 and DAC 2025).

**Industrial EDA Development – CASTEST** Sep. 2021 – Jan. 2025
Beijing, China

- Developed core algorithms and implementation for fault simulation and ISO 26262-oriented functional safety analysis for automotive chips.
- Worked on tool integration, multi-core performance optimization, requirements, and deployment for industrial users.

**RRAM-Based True Random Number Generator** Sep. 2020 – Jun. 2022
Undergraduate thesis, Peking University

- Designed an RRAM-based generator and evaluated randomness with the NIST SP 800-22 statistical test suite.

## Publications

† Equal contribution. ‡ Co-corresponding authors.

### Conference Papers

**CU-3D: A Scalable Legality-Aware RTL-to-GDS Flow for 3D-IC Physical Design**
**Mingjun Wang**, Guanqi Li, Yihan Wen, Lancheng Zou, Mingjun Li, Feng Gu, Fengrui Liu, Jianan Mu, Yuxuan Zhao, Tsung-Yi Ho, Huawei Li, Sung Kyu Lim, Bei Yu.
IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC) · Tokyo, Jan. 25–28, 2027. **Invited Paper**

**HOIST: A High-Order Interaction Surrogate for Sample-Efficient High-Sigma Yield Estimation in High-Dimensional Circuits**
Zhongru Xiong, **Mingjun Wang**‡, Yanfang Liu, Tsung-Yi Ho, Bei Yu, Binwu Zhu‡.
IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC) · Tokyo, Jan. 25–28, 2027. Accepted.

**Logic Resynthesis for Cross-Die Dependencies in Hybrid Bonded 3D ICs**
Fangzhou Liu, Yuhao Ji, Lancheng Zou, Jiawei Liu, **Mingjun Wang**, Gengjie Chen, Guannan Guo, Bei Yu.
IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC) · Tokyo, Jan. 25–28, 2027. Accepted.

**Distill3D: Cross-Stage Knowledge Distillation for Early-Stage 3D IC Timing Prediction**
**Mingjun Wang**†, Lancheng Zou†, Guanqi Li†, Feng Gu, Yuntao Lu, Tianmeng Yang, Boyu Han, Jianan Mu, Huawei Li, Sung Kyu Lim, Bei Yu.
International Conference on Computer-Aided Design (ICCAD), 2026. Accepted.

**CircuitNet 3.0: A Multi-Modal Dataset with Task-Oriented Augmentation for AI-Driven Circuit Design**
**Mingjun Wang**†, Yihan Wen†, Yuntao Lu, Fengrui Liu, Yuxiang Zhao, Boyu Han, Jianan Mu, Yibo Lin, Runsheng Wang, Bei Yu, Huawei Li.
International Conference on Learning Representations (ICLR), 2026.

**ATLAS: Asynchronous Topological Learning for Accurate FIT Prediction via Decoupled Graph Neural Networks**
Yicheng Liu†, **Mingjun Wang**†, Songwei Pei, Yuntao Lu, Zizhen Liu, Boyu Han, Huawei Li, Shangguang Wang.
Design Automation Conference (DAC), 2026.

**AnalogVerifier: A Neuro-Symbolic Framework for Analog Circuit Verification**
Yanfang Liu†, **Mingjun Wang**†, Peng Xu, Rongliang Fu, Bei Yu, Tsung-Yi Ho.
International Conference on Machine Learning (ICML), 2026.

**LibPilot: Knowledge-Constrained Dual-Agent Reasoning for Standard Cell Library Tuning**
Yanfang Liu, Zijin Cheng, **Mingjun Wang**, Rongliang Fu, Chao Wang, Bei Yu.
International Conference on Computer-Aided Design (ICCAD), 2026. Accepted.

**ChatFCM: Comprehension-Synthesis Decoupling for LLM-Based Functional Coverage Model Generation**
Yuan Pu, Yudong He, Zhenghao Chen, **Mingjun Wang**, Bingkun Yao, Zhaotan Lin, Qin Chen, Bing Li, Song Chen, Zhuolun He, Wenjian Yu, Yanfeng Li, Bei Yu.
International Conference on Computer-Aided Design (ICCAD), 2026. Accepted.

**MigSizer: Physical-Aware Sizing Improving Analog/RF Circuit Migration**
Yanfang Liu, Peng Xu, Yapeng Li, **Mingjun Wang**, Leilei Jin, Rongliang Fu, Tinghuan Chen, Wei Xing, Bei Yu, Tsung-Yi Ho.
IEEE International Conference on Computer Design (ICCD) · Hong Kong, Nov. 16–18, 2026. Accepted.

**LLM-Assisted Circuit Verification: A Comprehensive Survey**
Hongduo Liu, Yuntao Lu, **Mingjun Wang**, Xufeng Yao, Bei Yu.
Asia and South Pacific Design Automation Conference (ASP-DAC), 2026. **Invited Paper**

**Bridging Layout and RTL: Knowledge Distillation based Timing Prediction**
**Mingjun Wang**†, Yihan Wen†, Bin Sun, Jianan Mu, Juan Li, Xiaoyi Wang, Jing Justin Ye, Bei Yu, Huawei Li.
International Conference on Machine Learning (ICML), 2025. **Spotlight**

**MOSS: Multi-Modal Representation Learning on Sequential Circuits**
**Mingjun Wang**, Bin Sun, Jianan Mu, Feng Gu, Boyu Han, Tianmeng Yang, Xinyu Zhang, Silin Liu, Yihan Wen, Hui Wang, Jun Gao, Zhiteng Chao, Husheng Han, Zizhen Liu, Shengwen Liang, Jing Ye, Bei Yu, Xiaowei Li, Huawei Li.
Design Automation Conference (DAC), 2025.

**EPICS: Efficient Parallel Pattern Fault Simulation for Sequential Circuits via Strongly Connected Components**
**Mingjun Wang**, Hui Wang, Jianan Mu, Xinyu Zhang, Bin Sun, Yihan Wen, Zizhen Liu, Feng Gu, Jun Gao, Shengwen Liang, Jing Ye, Xiaowei Li, Huawei Li.
Design Automation Conference (DAC), 2025.

**ETPG: Efficient Transition Fault Simulation via Dual-Strategy Pattern Parallelism and Gate Restructuring**
**Mingjun Wang**, Hui Wang, Zizhen Liu, Feng Gu, Jianan Mu, Jiaping Tang, Jun Gao, Huawei Li, Jing Ye, Xiaowei Li.
Asia and South Pacific Design Automation Conference (ASP-DAC), 2025. **Best Paper Nomination**

**VIRTUAL: Vector-based Dynamic Power Estimation via Decoupled Multi-Modality Learning**
Yuntao Lu†, **Mingjun Wang**†, Yihan Wen, Boyu Han, Jianan Mu, Huawei Li, Bei Yu.
International Conference on Computer-Aided Design (ICCAD), 2025.

**DFTS: An Efficient Design-for-Test Flow for Scan Design**
Jun Gao, **Mingjun Wang**, Jiangwang Liu, Wenjie Li, Zizhen Liu, Jing Ye, Huawei Li.
International Symposium of EDA (ISEDA), 2025.

**Efficient Functional Safety Method for Gate-Level Fine-Grained Digital Circuits with ISO-26262**
**Mingjun Wang**, Hui Wang, Jianan Mu, Zizhen Liu, Jun Gao, Jing Ye, Huawei Li, Xiaowei Li.
International Test Conference in Asia (ITC-Asia), 2024.

**DDP-Fsim: Efficient and Scalable Fault Simulation for Deterministic Patterns with Two-Dimensional Parallelism**
Feng Gu, **Mingjun Wang**, Jianan Mu, Zizhen Liu, Jiaping Tang, Hui Wang, Yonghao Wang, Jing Ye, Huawei Li, Xiaowei Li.
International Conference on Computer-Aided Design (ICCAD), 2024. **Best Paper Nomination**

## Journal Papers

**DomSim: Hardware-Aware Hybrid Fault Simulation With Dominator Tree-Guided Partitioning**
**Mingjun Wang**†, Hui Wang†, Feng Gu, Zizhen Liu, Jianan Mu, Shengwen Liang, Zhongkai Yu, Zheng Liang, Jun Gao, Jiaping Tang, Jing Ye, Xiaowei Li, Bei Yu, Huawei Li.
IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 45(6): 2725–2738, 2026. Early Access: 2025.

**Bit-Compressed Concurrent Fault Simulation with Efficient Fault List**
Feng Gu, Zhiteng Chao, Hui Wang, Jianan Mu, **Mingjun Wang**, Jun Gao, Yonghao Wang, Jing Ye, Xiaowei Li, Huawei Li.
IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Early Access, 2026.

## Honors and Awards

- **Winner**, IEEE ICLAD-DAC GenAI Hackathon, Design Verification SLM Track, 2025.
- **Second Place**, EDAthon Programming Competition, IEEE CEDA, 2025.
- **Second Prize, Enterprise Track**, 8th China Graduate IC Innovation Contest, 2025.
- **Third Prize, Team Competition**, 8th China Graduate IC Innovation Contest, 2025.
- International Exchange Scholarship for Empowerment, ICT, CAS, 2025.
- **Director's Special Award**, ICT, CAS, 2024 (1st out of 400 students).
- **National Scholarship**, 2024 (Top 2%).
- **Special Prize**, LLM4EDA Summer School Hackathon, 2024 (1st out of 100 participants).
- Academic Endeavor Award, EDA Elite Challenge, 2024.
- First Prize Academic Scholarship, University of Chinese Academy of Sciences, 2024 (Top 15%).
- Merit Student, University of Chinese Academy of Sciences, 2023 and 2024.
- **First Prize**, EDA$^2$ Integrated Circuit Elite Challenge, Huawei HiSilicon, 2023 (1st out of 230 teams).
- Future Star, MOE–Huawei Intelligent Base Program, 2023.
- Cambricon Outstanding Student Award, 2023 (10 out of 300 students).
- Outstanding Contribution Award, Beijing 2022 Winter Olympics, PKU, 2022 (Top 10%).
- Peking University Social Work Award, 2019 and 2020.

## Technical Skills

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|---|---|
| **Languages** | Python, C++, SystemVerilog, Verilog, Tcl. |
| **EDA Tools** | Cadence Innovus; Synopsys PrimeTime, Design Compiler, VCS; Cadence Xcelium. |
| **AI / ML** | PyTorch, CUDA, PyTorch Geometric, TensorFlow. |
| **Development** | Linux, Git, LaTeX, reproducible experiment workflows. |

## Leadership and Service

- **President, Student Science and Technology Association**, ICT, CAS, Sep. 2023 – Dec. 2024. Organized 16 science popularization sessions.
- **Deputy Secretary-General, Student Activity Center**, Peking University, Oct. 2020 – Apr. 2022. Organized university events and volunteer recruitment.
- **Volunteer Leader, Beijing 2022 Winter Olympics**, International Olympic Committee, Oct. 2021 – Mar. 2022. Served as an Olympic Family Assistant.